Cryptographic Computing
Theory Assignment

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About this assignment. The assignment has three questions,[1] of which you should attempt *at least two*. In order to pass, your answers need not be 100% correct, but they should show a reasonable amount of effort with sufficient understanding of the security definitions (see Week 2-3 notes and the book, linked on Brightspace) and how to apply them to analyze concrete protocols.

A security proof consists of (1) a precise description of the simulator, i.e., which inputs it receives and how it computes every message in the simulated view, and (2) an argument that the simulated view and outputs are distributed in the same way as the real ones, or are indistinguishable from them. You may use any result proved in the lecture notes, as long as you say which one.

Question 1: Active Security of BeDOZa with MACs

In week 3 we added information-theoretic MACs to the arithmetic BeDOZa protocol, but we only sketched why the resulting protocol is secure. In this question you will write a proof for a small circuit.

Let p be a large prime. Alice has input $x \in \mathbb{Z}_p$, Bob has input $y \in \mathbb{Z}_p$, and they want to compute

$$f(x, y) = (x \cdot y,\ \perp),$$

i.e., only Alice learns the product. We use the authenticated representation $[x]$, the subprotocols OpenTo, OpenToAll, addition, addition with a constant and multiplication by a constant from the notes. Recall that $\mathsf{Tag}((\alpha, \beta), x) = \alpha x + \beta$, that Alice's global key α_A and Bob's global key α_B are the same for every wire, and that in OpenToAll we let Alice send her share and tag first and then Bob send his. If a MAC check fails, Alice aborts and outputs $f(x, y_0)$ for a fixed default input y_0, and Bob aborts and outputs nothing.

The protocol in the dealer model is the following:

Dealer: The trusted dealer samples $\alpha_A, \alpha_B \leftarrow \mathbb{Z}_p$ and outputs to Alice and Bob authenticated sharings $[r], [s], [u], [v], [w]$ of uniformly random $r, s, u, v \in \mathbb{Z}_p$ and $w = uv$. Every tag is computed with a fresh, uniformly random local key β.

Alice's input: The parties run $(r, \perp) \leftarrow \text{OpenTo}(A, [r])$. Alice sends $\delta_x = x - r$ to Bob, and both compute $[x] = [r] + \delta_x$.

Bob's input: The parties run $(\perp, s) \leftarrow \text{OpenTo}(B, [s])$. Bob sends $\delta_y = y - s$ to Alice, and both compute $[y] = [s] + \delta_y$.

Multiplication: The parties compute $[d] = [x] + [u]$ and $[e] = [y] + [v]$, run $d \leftarrow \text{OpenToAll}([d])$ and $e \leftarrow \text{OpenToAll}([e])$, and compute $[z] = [w] + e[x] + d[y] - de$.

Output: The parties run $(z, \perp) \leftarrow \text{OpenTo}(A, [z])$, and Alice outputs z.

Before writing the proof, you may want to remind yourself how the MAC verification works and convince yourself that the protocol is correct.

[1] These have been adapted from exercises in the lecture notes which were not covered in class.

1. Write down the view of Bob: his randomness and every message he receives. Also list every message that Bob sends to Alice, and for each one say which value an honest Bob would send.

2. Let Bob be actively corrupted. Describe a simulator $\mathcal{S}_B$ which interacts with the corrupted Bob. Explain (i) how $\mathcal{S}_B$ extracts an input y^* from Bob, (ii) how it simulates the messages of the honest Alice and of the dealer *without knowing* x, and (iii) how it decides which input to send to the ideal functionality.

3. Show that the view of the corrupted Bob is distributed *identically* in the real and in the simulated execution. (Note: the *view on its own* doesn't include Alice's input or output, so this does not yet prove security)

4. Argue that for all inputs $x \in \mathbb{Z}_p$,[2] the joint distributions of $(x, \mathsf{view}_B, z_A)$ in the real and in the simulated execution have statistical distance at most c/p, where view_B is the view of the corrupted Bob and z_A is the output of Alice, for a small constant c which you should find. Identify the "bad event" on which the two executions differ, and explain why the probability of this event is small.

Question 2: Passive Security of IKNP against a Corrupted Receiver

Recall the IKNP OT extension protocol from Week 4. As in the notes, Alice is the *receiver* of the N extended OTs, with choice bits $r \in \{0,1\}^N$, and Bob is the *sender*, with message pairs $(m_{j,0}, m_{j,1})_{j\in[N]}$ where $m_{j,b} \in \{0,1\}^\ell$. We analyse the protocol in the OT-hybrid model: the λ base OTs are ideal $\binom{2}{1}$-OTs on N-bit strings, so Alice (who is the *sender* in the base OTs) learns nothing from them.

The hash function $H : [N] \times \{0,1\}^\lambda \to \{0,1\}^\ell$ is *correlation robust*: for uniformly random $s, t_1, \ldots, t_N \in \{0,1\}^\lambda$ and uniformly random $u_1, \ldots, u_N \in \{0,1\}^\ell$,

$$(t_1, \ldots, t_N,\ H(1, t_1 \oplus s), \ldots, H(N, t_N \oplus s)) \approx (t_1, \ldots, t_N,\ u_1, \ldots, u_N)\,.$$

Note that the only information that isn't given to the distinguisher is the secret correlation s.

1. Write down precisely the view view_A of a passively corrupted Alice in the protocol, and the outputs of both parties.

2. Describe a simulator $\mathcal{S}_A$ which on input 1^λ, Alice's input r and her output $(m_{j,r_j})_{j\in[N]}$ produces a simulated view.

3. Show that for all inputs, the joint distribution of the simulated view and the inputs/outputs is computationally indistinguishable from the joint distribution of the real view and the inputs/outputs. Do this by (a) Writing down how what the distinguisher receives differs between the real/ideal worlds, and then (b) Showing that a distinguisher $\mathcal{D}$ with advantage ϵ can be turned into a distinguisher $\mathcal{D}'$ with advantage ϵ for the correlation robustness of H. Be explicit about how $\mathcal{D}'$ uses its challenge, and check that $\mathcal{D}'$ gives $\mathcal{D}$ exactly the real distribution in one case and exactly the ideal distribution in the other.

Question 3: Oblivious Transfer from RSA

In Week 4 we built OT from any public-key encryption scheme in which public keys can be sampled obliviously. This can be done with ElGamal, but not RSA: we don't know any way to sample a modulus $N = pq$ without learning the factorisation. However, RSA has a different property: given a public key $pk = (n, e)$, it is easy to generate "ciphertexts" that look like encryptions of random messages, without knowing which messages they encrypt. Indeed, an RSA encryption $c = m^e \bmod n$ of a uniformly random $m \in \mathbb{Z}_n^*$ is a

[2] Note that we have omitted Bob's input y, since the corrupt Bob may just ignore his prescribed input.

uniformly random element of $\mathbb{Z}_n^*$, and such elements can be sampled directly. In this question you will use this property to build an OT protocol. Use the same roles as in the notes: Alice is the receiver with choice bit b, and Bob is the sender with messages m_0, m_1.

1. Formalise the property "it is possible to generate ciphertexts which look like encryptions of random messages, without knowing their plaintexts" by defining an invertible algorithm OEnc, following the blueprint of OGen. Define the syntax of OEnc and of its inverse OEnc^{-1}, and state the properties they must satisfy, including a security property saying that the plaintext of $\mathsf{OEnc}(pk; \rho)$ stays hidden even given the randomness ρ.

2. Construct a $\binom{2}{1}$-O protocol from RSA (or from any scheme with an OEnc algorithm as in part 1), and show that it is correct.

3. Sketch a security proof for your protocol against passive adversaries, by writing down a simulator for each of the two parties. Argue informally why it is secure (you don't need to do the reduction).

4. Show that your protocol is *not* secure if Alice is actively corrupted.

($\star$, optional) Show that an actively corrupted Bob can learn information about b by choosing his public key maliciously.

Hint 1: Gur cebgbpby unf 3 ebhaqf.

Hint 2: Va gur svefg ebhaq, Obo fraqf na EFN choyvp xrl gb Nyvpr. Gura, Nyvpr pna hfr OEnc (fvzvyneyl gb ubj fur hfrq OGen).

Hint 3: Sbe cneg ($\star$): jung unccraf gb gur znc $x \mapsto x^e \bmod n$ vs $\gcd(e, \phi(n)) \neq 1$?